Lesson 33: Normal-Form Games and Pure Nash Equilibrium

AI/ML Mathematics

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1 Motivation: Why Game Theory for Multi-Agent RL

In single-agent reinforcement learning, the environment is modeled as a Markov decision process whose transition kernel does not *adapt* to the agent's behavior. Multi-agent reinforcement learning (MARL) breaks this assumption: when two or more learning agents share an environment, each agent's effective "environment" includes the other agents, whose policies change as they learn. The resulting non-stationarity means that standard convergence guarantees fail. To reason about what agents *should* converge to, we need a mathematical framework for strategic interaction — and that framework is game theory.

RL/ML Connection

Every MARL algorithm implicitly targets a game-theoretic solution concept:

- **Independent Q-learning**: each agent ignores others; if it converges, the result is a Nash equilibrium of the stage game.
- **Minimax-Q** [Shoham and Leyton-Brown, 2008]: targets the minimax value in zero-sum stochastic games, applying von Neumann's minimax theorem.
- **Nash-Q**: computes Nash equilibria of stage games at each step.
- **MAPPO, QMIX**: cooperative algorithms whose analysis relies on common-payoff game structure.

Without understanding Nash equilibria and dominance, one cannot state what these algorithms compute, let alone prove convergence.

This lesson develops the foundational concepts: normal-form games, dominance, iterated elimination, best responses, rationalizability, and pure strategy Nash equilibrium. Four classical games serve as running examples.

Prerequisites. Finite probability spaces, basic linear algebra (convex sets, hyperplane separation), and elementary real analysis.

2 Normal-Form Games

Definition 2.1 (Normal-Form Game)**.** A *normal-form game* (or *strategic-form game*) is a tuple $\Gamma = (N, (S_i)_{i\in N}, (u_i)_{i\in N})$ where:

(i) $N = \{1, 2, \ldots, n\}$ is a finite set of *players*.

(ii) For each $i \in N$, S_i is a non-empty set of *strategies* (or *actions*) available to player i. The set of *strategy profiles* is $S = \prod_{i\in N} S_i$.

(iii) For each $i \in N$, $u_i \colon S \to \mathbb{R}$ is player i's *payoff function* (or *utility function*).

The game is *finite* if every S_i is a finite set.

Remark 2.2 (Notation). For a strategy profile $s = (s_1, \ldots, s_n) \in S$, we write $s_{-i} = (s_1, \ldots, s_{i-1}, s_{i+1}, \ldots, s_n) \in S_{-i} := \prod_{j\neq i} S_j$ and use $s = (s_i, s_{-i})$ to emphasize player i's component.

Definition 2.3 (Bimatrix Game)**.** A *bimatrix game* is a two-player finite normal-form game presented as a pair of matrices $(A, B) \in \mathbb{R}^{m\times k} \times \mathbb{R}^{m\times k}$, where player 1 has m strategies, player 2 has k strategies, $A_{jl} = u_1(j, l)$, and $B_{jl} = u_2(j, l)$. If $B = -A$, the game is *zero-sum*.

Example 2.4 (Prisoner's Dilemma)**.** Two players simultaneously choose to *Cooperate* (C) or *Defect* (D). The payoff bimatrix is:

	C	D
C	$(3, 3)$	$(0, 5)$
D	$(5, 0)$	$(1, 1)$

Each entry (a, b) gives (row player's payoff, column player's payoff). Regardless of the opponent's choice, Defect yields a strictly higher payoff, making D a strictly dominant strategy for both players.

Example 2.5 (Battle of the Sexes)**.** Players 1 and 2 prefer different activities — *Opera* (O) and *Football* (F) — but both prefer coordination to miscoordination:

	O	F
O	$(3, 1)$	$(0, 0)$
F	$(0, 0)$	$(1, 3)$

Neither player has a dominant strategy. Both (O, O) and (F, F) will turn out to be pure Nash equilibria (Section 5).

Example 2.6 (Matching Pennies)**.** A zero-sum game: each player shows *Heads* (H) or *Tails* (T). Player 1 wins if the choices match; player 2 wins if they differ:

	H	T
H	$(1, -1)$	$(-1, 1)$
T	$(-1, 1)$	$(1, -1)$

This game has no pure Nash equilibrium (Proposition 6.5), motivating the extension to mixed strategies in Lesson 34.

Example 2.7 (Stag Hunt)**.** Two hunters choose to hunt a *Stag* (S, requires cooperation) or a *Hare* (H, safe but less rewarding):

	S	H
S	$(4, 4)$	$(0, 3)$
H	$(3, 0)$	$(3, 3)$

Both (S, S) and (H, H) are pure Nash equilibria. The equilibrium (S, S) is *payoff-dominant* (higher payoffs for both players), while (H, H) is *risk-dominant* (safer against uncertainty about the opponent's choice).

RL/ML Connection

The Stag Hunt captures a fundamental tension in cooperative MARL: agents can achieve higher collective reward through coordination, but each agent bears risk if the others fail to coordinate. This is precisely the challenge faced by centralized-training decentralized-execution (CTDE) algorithms such as QMIX and MAPPO, where the training procedure must guide agents toward the payoff-dominant equilibrium despite the safety of the risk-dominant one.

3 Dominance and Iterated Elimination

Definition 3.1 (Strict and Weak Dominance)**.** Let $\Gamma = (N, (S_i), (u_i))$ be a normal-form game and fix player i.

(a) A strategy $s_i \in S_i$ *strictly dominates* $s_i' \in S_i$ if

$$u_i(s_i, s_{-i}) > u_i(s_i', s_{-i}) \quad \text{for all } s_{-i} \in S_{-i}.$$

(b) A strategy s_i *weakly dominates* s_i' if

$$u_i(s_i, s_{-i}) \geq u_i(s_i', s_{-i}) \quad \text{for all } s_{-i} \in S_{-i},$$

with strict inequality for at least one s_{-i}.

A strategy is *strictly dominated* if some other strategy strictly dominates it. A strategy is a *strictly dominant strategy* if it strictly dominates every other strategy of that player.

Example 3.2 (Dominance in the Prisoner's Dilemma)**.** In Example 2.4, for player 1: $u_1(D, C) = 5 > 3 = u_1(C, C)$ and $u_1(D, D) = 1 > 0 = u_1(C, D)$. Hence D strictly dominates C for player 1. By symmetry, D strictly dominates C for player 2 as well. Both players have a strictly dominant strategy.

Proposition 3.3 (Dominated Strategies Are Never Best Responses)**.** *If s_i' is strictly dominated by s_i, then s_i' is never a best response: for every $s_{-i} \in S_{-i}$, $s_i' \notin \arg\max_{t_i \in S_i} u_i(t_i, s_{-i})$.*

Proof. Fix any $s_{-i} \in S_{-i}$. Since s_i strictly dominates s_i', we have $u_i(s_i, s_{-i}) > u_i(s_i', s_{-i})$. Therefore s_i' does not achieve the maximum of $u_i(\cdot, s_{-i})$ over S_i, so $s_i' \notin \arg\max_{t_i \in S_i} u_i(t_i, s_{-i})$. □

Warning

The converse of Proposition 3.3 is false in general for pure strategies: a strategy can fail to be a best response to any pure opponent profile without being strictly dominated by any single pure strategy. However, in two-player games with finite strategy sets, a pure strategy that is never a best response is strictly dominated by some *mixed* strategy (a consequence of the supporting hyperplane theorem).

Definition 3.4 (Iterated Elimination of Strictly Dominated Strategies)**.** *Iterated elimination of strictly dominated strategies* (IESDS) proceeds as follows. Set $S_i^0 = S_i$ for all $i \in N$. For $k = 0, 1, 2, \ldots$, define

$$S_i^{k+1} = \left\{ s_i \in S_i^k : \text{no } s_i' \in S_i^k \text{ satisfies } u_i(s_i', s_{-i}) > u_i(s_i, s_{-i}) \text{ for all } s_{-i} \in S_{-i}^k \right\}.$$

The set of *surviving strategies* for player i is $S_i^\infty = \bigcap_{k=0}^\infty S_i^k$. In a finite game, this intersection stabilizes after finitely many rounds.

Remark 3.5. Definition 3.4 eliminates *all* strictly dominated strategies simultaneously at each round. One could instead eliminate strategies one at a time, choosing different players and strategies in different orders. Theorem 3.7 shows that the final surviving set is the same regardless of the elimination order.

Example 3.6 (IESDS in the Prisoner's Dilemma)**.** In the Prisoner's Dilemma (Example 2.4), C is strictly dominated by D for both players. Round 0 eliminates C for both, leaving $S_1^1 = S_2^1 = \{D\}$. The unique surviving profile is (D, D).

Theorem 3.7 (Order Independence of IESDS)**.** *In a finite normal-form game, the set of strategy profiles surviving IESDS is independent of the order in which strategies are eliminated.*

Proof. We show that any strategy eliminated under one elimination order is eventually eliminated under every elimination order. The proof proceeds by strong induction on the round of elimination.

Let E and E' denote two arbitrary elimination orders. Write $S_i^k(E)$ for the surviving strategies of player i after round k under order E, and similarly $S_i^k(E')$ for order E'. Write $S_i^\infty(E)$ and $S_i^\infty(E')$ for the respective final surviving sets.

We prove by strong induction on k that if a strategy s_i^* is eliminated at round k under order E, then $s_i^* \notin S_i^\infty(E')$.

Base case ($k = 0$). If s_i^* is eliminated at round 0 under E, then there exists $s_i' \in S_i$ with $u_i(s_i', s_{-i}) > u_i(s_i^*, s_{-i})$ for all $s_{-i} \in S_{-i}$. Since $S_{-i}^\infty(E') \subseteq S_{-i}$, the domination holds on $S_{-i}^\infty(E')$. If $s_i' \in S_i^\infty(E')$, then $s_i^* \notin S_i^\infty(E')$ directly. Otherwise, following the chain of dominators from

s_i' to some $\hat{s}_i \in S_i^\infty(E')$ (the chain terminates since S_i is finite), we get $u_i(\hat{s}_i, s_{-i}) > u_i(s_i^*, s_{-i})$ on $S_{-i}^\infty(E')$, so again $s_i^* \notin S_i^\infty(E')$.

Inductive step. Assume the claim holds for all strategies eliminated in rounds $0, 1, \ldots, k-1$ under any elimination order. Suppose s_i^* is eliminated at round k under order E. Then there exists $s_i' \in S_i^{k-1}(E)$ such that

$$u_i(s_i', s_{-i}) > u_i(s_i^*, s_{-i}) \quad \text{for all } s_{-i} \in S_{-i}^{k-1}(E). \tag{1}$$

Claim: $S_{-i}^\infty(E') \subseteq S_{-i}^{k-1}(E)$.

To prove the claim, take any $s_j \in S_j^\infty(E')$ for $j \neq i$. Suppose for contradiction that $s_j \notin S_j^{k-1}(E)$. Then s_j was eliminated at some round $l < k$ under order E. By the inductive hypothesis, $s_j \notin S_j^\infty(E')$, contradicting $s_j \in S_j^\infty(E')$. Hence $s_j \in S_j^{k-1}(E)$, establishing the claim.

By the claim, the strict inequality (1) holds for all $s_{-i} \in S_{-i}^\infty(E')$. We now consider two cases.

Case 1: $s_i' \in S_i^\infty(E')$. Then s_i^* is strictly dominated by s_i' on the surviving set $S_{-i}^\infty(E')$, so $s_i^* \notin S_i^\infty(E')$.

Case 2: $s_i' \notin S_i^\infty(E')$. Then s_i' was eliminated at some round m under E', meaning there exists $s_i'' \in S_i^{m-1}(E')$ that strictly dominates s_i' on $S_{-i}^{m-1}(E')$. Repeating this chain, since S_i is finite, we eventually reach some $\hat{s}_i \in S_i^\infty(E')$ that strictly dominates s_i' on $S_{-i}^\infty(E')$. Combining with (1): for all $s_{-i} \in S_{-i}^\infty(E')$, $u_i(\hat{s}_i, s_{-i}) > u_i(s_i', s_{-i}) > u_i(s_i^*, s_{-i})$. Hence s_i^* is strictly dominated by $\hat{s}_i$ on the surviving set, so $s_i^* \notin S_i^\infty(E')$.

By induction, $S_i^\infty(E) \supseteq S_i^\infty(E')$ for all i. Swapping the roles of E and E' gives the reverse inclusion, so $S_i^\infty(E) = S_i^\infty(E')$. □

Key Result

The surviving set under IESDS is a canonical object of the game: it does not depend on procedural choices. In a game solvable by IESDS (where the surviving set is a singleton), IESDS yields the unique Nash equilibrium.

4 Best Response and Rationalizability

Definition 4.1 (Best Response Correspondence)**.** Player i's *best response correspondence* is the set-valued map $\mathrm{BR}_i \colon S_{-i} \rightrightarrows S_i$ defined by

$$\mathrm{BR}_i(s_{-i}) = \arg\max_{s_i \in S_i} u_i(s_i, s_{-i}).$$

In a finite game, $\mathrm{BR}_i(s_{-i})$ is non-empty for every $s_{-i} \in S_{-i}$, since a finite set always attains its maximum.

Example 4.2 (Best Responses in Battle of the Sexes)**.** In Example 2.5, compute player 1's best responses:

- If player 2 plays O: $u_1(O, O) = 3 > 0 = u_1(F, O)$, so $\mathrm{BR}_1(O) = \{O\}$.
- If player 2 plays F: $u_1(F, F) = 1 > 0 = u_1(O, F)$, so $\mathrm{BR}_1(F) = \{F\}$.

By symmetry (with swapped preferences), $\mathrm{BR}_2(O) = \{O\}$ and $\mathrm{BR}_2(F) = \{F\}$. The profiles where both players play a mutual best response are (O, O) and (F, F).

Definition 4.3 (Rationalizable Strategies)**.** A strategy s_i is *rationalizable* if it survives iterated elimination of strategies that are never best responses. Formally, set $R_i^0 = S_i$ and define

$$R_i^{k+1} = \left\{ s_i \in R_i^k : \exists\, s_{-i} \in R_{-i}^k \text{ such that } s_i \in \mathrm{BR}_i(s_{-i}) \right\}.$$

The set of *rationalizable strategies* is $R_i^\infty = \bigcap_{k=0}^\infty R_i^k$.

Remark 4.4. Rationalizability captures "common knowledge of rationality": a strategy is rationalizable if it is a best response to some belief about the opponents, where that belief is consistent with the opponents being rational, and so on ad infinitum.

Proposition 4.5 (IESDS and Rationalizability in Two-Player Games)**.** *In a finite two-player game, the set of rationalizable strategies coincides with the set of strategies surviving IESDS: $R_i^\infty = S_i^\infty$ for $i = 1, 2$.*

Proof. We establish both inclusions.

($S_i^\infty \supseteq R_i^\infty$). Every strictly dominated strategy is never a best response (Proposition 3.3). Hence every strategy eliminated in IESDS is also eliminated in the rationalizability procedure, giving $R_i^k \subseteq S_i^k$ for all k by induction. Taking intersections, $R_i^\infty \subseteq S_i^\infty$.

($S_i^\infty \subseteq R_i^\infty$). We show the contrapositive: if $s_i \notin R_i^\infty$, then $s_i \notin S_i^\infty$. Suppose s_i is eliminated at round k in the rationalizability procedure, meaning s_i is not a best response to any $s_{-i} \in R_{-i}^{k-1}$.

In a two-player game, this means: for every $s_j \in R_j^{k-1}$ (where $j \neq i$), there exists $s_i'(s_j) \in R_i^{k-1}$ with $u_i(s_i'(s_j), s_j) > u_i(s_i, s_j)$. We claim that s_i is strictly dominated on R_j^{k-1} (possibly by a mixed strategy). Indeed, consider the payoff vectors $v = (u_i(s_i, s_j))_{s_j \in R_j^{k-1}} \in \mathbb{R}^{|R_j^{k-1}|}$ and the convex set $W = \{(u_i(s_i', s_j))_{s_j \in R_j^{k-1}} : s_i' \in R_i^{k-1}\}$. Since s_i is never a best response on R_j^{k-1}, by the supporting hyperplane theorem applied to the convex hull of payoff vectors, there exists a convex combination (i.e., a mixed strategy) over R_i^{k-1} that strictly dominates s_i on R_j^{k-1}.

For pure-strategy IESDS where we also allow elimination by mixed-strategy domination, this gives $s_i \notin S_i^\infty$. In finite two-player games, elimination by pure or mixed strict domination yields the same surviving set [Osborne and Rubinstein, 1994], completing the proof. □

Warning

For games with three or more players, the equivalence between IESDS and rationalizability can fail when restricting to pure-strategy domination. Rationalizability may be strictly coarser than pure-strategy IESDS survival for $n \geq 3$. The equivalence is restored when domination by mixed strategies is permitted [Fudenberg and Tirole, 1991].

5 Pure Strategy Nash Equilibrium

Definition 5.1 (Pure Strategy Nash Equilibrium)**.** A strategy profile $s^* = (s_1^*, \ldots, s_n^*) \in S$ is a *pure strategy Nash equilibrium* (NE) if no player can unilaterally improve their payoff:

$$u_i(s_i^*, s_{-i}^*) \geq u_i(s_i, s_{-i}^*) \quad \text{for all } s_i \in S_i, \text{ for all } i \in N.$$

Proposition 5.2 (Nash Equilibrium as Mutual Best Response)**.** *A strategy profile s^* is a Nash equilibrium if and only if $s_i^* \in \mathrm{BR}_i(s_{-i}^*)$ for every $i \in N$.*

Proof. By Definition 5.1, s^* is a NE if and only if $u_i(s_i^*, s_{-i}^*) \geq u_i(s_i, s_{-i}^*)$ for all $s_i \in S_i$ and all $i \in N$. By Definition 4.1, this is precisely $s_i^* \in \arg\max_{s_i \in S_i} u_i(s_i, s_{-i}^*) = \mathrm{BR}_i(s_{-i}^*)$ for all i. □

Key Result

Proposition 5.2 gives a practical method for finding all pure Nash equilibria in finite games: for each player and each opponent profile, compute the best response set, then identify profiles where every player's strategy is a best response. This "underline the best responses" method works for any bimatrix game.

Proposition 5.3 (Dominant Strategy Equilibrium Implies Nash Equilibrium)**.** *If s_i^* is a strictly dominant strategy for every player $i \in N$, then $(s_1^*, \ldots, s_n^*)$ is the unique Nash equilibrium.*

Proof. Since s_i^* strictly dominates every other strategy of player i, we have $u_i(s_i^*, s_{-i}) > u_i(s_i, s_{-i})$ for all $s_i \neq s_i^*$ and all $s_{-i} \in S_{-i}$. In particular, for $s_{-i} = s_{-i}^*$, this gives $s_i^* \in \mathrm{BR}_i(s_{-i}^*)$ for all i, so s^* is a NE by Proposition 5.2.

For uniqueness, suppose $\hat{s} \neq s^*$ is a NE. Then some $\hat{s}_i \neq s_i^*$, but $u_i(s_i^*, \hat{s}_{-i}) > u_i(\hat{s}_i, \hat{s}_{-i})$ by strict dominance, contradicting $\hat{s}_i \in \mathrm{BR}_i(\hat{s}_{-i})$. □

6 Analysis of Classic Games

We systematically analyze the four classic games from Section 2, identifying all pure Nash equilibria.

6.1 Prisoner's Dilemma

Proposition 6.1. *The Prisoner's Dilemma (Example 2.4) has a unique pure Nash equilibrium at (D, D) with payoffs $(1, 1)$.*

Proof. By Example 3.2, D is a strictly dominant strategy for both players. By Proposition 5.3, (D, D) is the unique Nash equilibrium. □

Remark 6.2 (Individual vs. Collective Rationality). The profile (C, C) yields payoffs $(3, 3)$, Pareto-dominating the NE (D, D) with payoffs $(1, 1)$. Yet rational players cannot sustain cooperation in a one-shot game. This tension is central to MARL: independent learners converge to the suboptimal NE unless the interaction is repeated (Lesson 36, folk theorems) or agents can commit to contracts.

6.2 Battle of the Sexes

Proposition 6.3. *The Battle of the Sexes (Example 2.5) has exactly two pure Nash equilibria: (O, O) with payoffs $(3, 1)$ and (F, F) with payoffs $(1, 3)$.*

Proof. From Example 4.2, $\mathrm{BR}_1(O) = \{O\}$, $\mathrm{BR}_1(F) = \{F\}$, $\mathrm{BR}_2(O) = \{O\}$, and $\mathrm{BR}_2(F) = \{F\}$. Of the four profiles, only (O, O) and (F, F) satisfy $s_i^* \in \mathrm{BR}_i(s_{-i}^*)$ for both players. At (O, F), player 1 deviates ($O \notin \mathrm{BR}_1(F)$); at (F, O), likewise. □

Remark 6.4 (Equilibrium Selection). The Battle of the Sexes illustrates the *equilibrium selection problem*: the theory predicts multiple equilibria but does not say which will be played. In MARL, independent learners may converge to different equilibria or cycle. Correlated equilibrium (Lesson 34) addresses this ambiguity.

6.3 Matching Pennies

Proposition 6.5 (Matching Pennies Has No Pure NE)**.** *The Matching Pennies game (Example 2.6) has no pure strategy Nash equilibrium.*

Proof. Player 1 wants to match and player 2 wants to mismatch, so their best responses oppose each other: $\mathrm{BR}_1(H) = \{H\}$, $\mathrm{BR}_1(T) = \{T\}$, $\mathrm{BR}_2(H) = \{T\}$, $\mathrm{BR}_2(T) = \{H\}$. At every profile, exactly one player has a profitable deviation:

- (H, H): player 2 deviates ($H \notin \mathrm{BR}_2(H)$).
- (H, T): player 1 deviates ($H \notin \mathrm{BR}_1(T)$).
- (T, H): player 1 deviates ($T \notin \mathrm{BR}_1(H)$).
- (T, T): player 2 deviates ($T \notin \mathrm{BR}_2(T)$).

No profile is a mutual best response, so no pure NE exists. □

Remark 6.6. The non-existence of pure NE reflects the adversarial structure of zero-sum games. Nash's theorem [Nash, 1950] guarantees a *mixed* NE (each player randomizes uniformly), which is the subject of Lesson 34. In competitive MARL, the best-response cycling $H \to T \to H \to \cdots$ is a simple instance of the oscillation that plagues naive independent learning; algorithms like minimax-Q and self-play converge to mixed equilibria instead.

6.4 Stag Hunt

Proposition 6.7. *The Stag Hunt (Example 2.7) has exactly two pure Nash equilibria:* (S, S) *with payoffs* $(4, 4)$ *and* (H, H) *with payoffs* $(3, 3)$.

Proof. Best responses: $\mathrm{BR}_1(S) = \{S\}$ (since $4 > 3$), $\mathrm{BR}_1(H) = \{H\}$ (since $3 > 0$), and by symmetry $\mathrm{BR}_2(S) = \{S\}$, $\mathrm{BR}_2(H) = \{H\}$. Only the diagonal profiles (S, S) and (H, H) satisfy mutual best response; at (S, H) and (H, S), the Stag-hunter deviates. □

Remark 6.8 (Payoff Dominance vs. Risk Dominance). The equilibrium (S, S) is *payoff-dominant* (both players prefer it), while (H, H) is *risk-dominant* [Fudenberg and Tirole, 1991]: S is a best response only when $p > 3/4$, where p is the believed probability the opponent plays S. In MARL, this tension determines whether independent learners converge to the Pareto-optimal or the risk-dominant equilibrium.

7 Exercises

Exercise 7.1 (Three-Player Dominance)**.** Consider a three-player game where each player chooses from $\{A, B\}$. Player 1's payoff is $u_1(s_1, s_2, s_3) = \mathbf{1}[s_1 = A] \cdot (1 + \mathbf{1}[s_2 = A] + \mathbf{1}[s_3 = A])$. That is, player 1 gets 0 from B and gets 1, 2, or 3 from A depending on how many others play A.

(a) Show that B is strictly dominated by A for player 1.

(b) Suppose u_2 and u_3 are defined analogously. Find all pure Nash equilibria.

(c) Apply IESDS and verify that the surviving set equals the set of Nash equilibria.

Exercise 7.2 (Best Response Analysis)**.** Consider the bimatrix game:

	L	C	R
T	$(2, 1)$	$(0, 2)$	$(1, 0)$
M	$(1, 2)$	$(3, 1)$	$(0, 0)$
B	$(0, 0)$	$(1, 3)$	$(2, 1)$

(a) Compute $\mathrm{BR}_1(s_2)$ for each $s_2 \in \{L, C, R\}$ and $\mathrm{BR}_2(s_1)$ for each $s_1 \in \{T, M, B\}$.

(b) Identify all pure Nash equilibria using the mutual best response characterization (Proposition 5.2).

(c) Apply IESDS. Does IESDS alone determine the equilibrium?

Exercise 7.3 (IESDS Order Independence)**.** Consider the following 3×3 bimatrix game:

	L	C	R
T	$(3, 1)$	$(0, 2)$	$(1, 0)$
M	$(1, 0)$	$(2, 3)$	$(0, 1)$
B	$(0, 2)$	$(1, 1)$	$(3, 0)$

(a) Perform IESDS by first eliminating dominated strategies for player 1, then player 2.

(b) Perform IESDS in the reverse order: first player 2, then player 1.

(c) Verify that both orders yield the same surviving set, confirming Theorem 3.7.

Exercise 7.4 (Coordination Game Design)**.** Design a symmetric 3×3 bimatrix game (i.e., $B = A^T$) with exactly three pure NE (all on the diagonal) and no strictly dominated strategies. Prove both properties hold.

Exercise 7.5 (Nash Equilibrium and IESDS Survival)**.** Prove that every Nash equilibrium of a finite game survives IESDS: if s^* is a pure NE, then $s_i^* \in S_i^\infty$ for all $i \in N$. *Hint:* Induction on the elimination round.

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